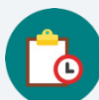
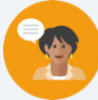


13.1 Simple Interest

Lesson Introduction

Lesson Introduction				
 Benchmarks	MA.912.FL.3.2 Solve real-world problems involving simple, compound, and continuously compounded interest. MA.912.FL.3.4 Explain the relationship between simple interest and linear growth. Explain the relationship between compound interest and exponential growth and the relationship between continuously compounded interest and exponential growth. MA.912.F.1.8 Determine whether a linear, quadratic, or exponential function best models a given real-world situation.		 Learning Targets	- I can explain the relationship between simple interest and linear growth. - I can solve real-world problems involving simple interest.
 Benchmark Coherence	Building On	Working Toward		
	N/A	MA.912.FL.3.1 Compare simple, compound, and continuously compounded interest over time. MA.912.FL.3.4 Goes beyond clarification where exponential growth is limited to compound interest.		
 Essential Questions	- What is the relationship between simple interest and linear growth? - How are real-world simple interest problems solved?			
 Lesson Narrative	In this lesson, students are introduced to the concept of simple interest. Students will begin by learning this concept and connecting it to what they have learned about linear growth, including key features of a linear function and their interpretation in each context. They will transition to calculating simple interest given a real-world problem both in a Guided Practice and Your Turn!			
 Component Summary	13.1.1 Warm-Up (~5 min.)	Earth's surface speed (<i>SE p. 235</i>)	13.1.3 Guided Practice (10-15 min.)	Practice calculating simple interest (<i>SE p. 239</i>)
	13.1.2 Guided Instruction (10-15 min.)	Introduction to simple interest and the connection to linear functions (<i>SE p. 236</i>)	13.1.4 Your Turn! (10-15 min.)	Independent practice with simple interest (<i>SE p. 240</i>)
	13.1.5 Wrap-Up (~5 min.)	SELFIE reflection (<i>SE p. 240</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Simple interest - Certificate of deposit (CD) <u>Additional Terms:</u> N/A		N/A	
			Additional Preparation (Optional) Additional information about CD savings accounts	

 Teacher Tips	Common Misconceptions - Students may not be able to identify key features. - Students may misinterpret the key features. - Students may write the percentage incorrectly.	Things to Consider During 13.1.2 Guided Instruction, students are learning the new concept of simple interest. Activate prior knowledge from linear functions, their graphs, and key features to give students opportunity to make meaningful connections to previous content.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect 13.1.2 Guided Instruction introduces students to the concept of simple interest. While this concept is new, it is designed to reinforce skills learned in prior units with linear functions and graphs. During this component, allow students to hear your thinking out loud as you connect the prior learning to the current learning.	MA.K12.MTR.5.1: Patterns and Structure One of the purposes of this lesson is to connect a new concept to what students have learned in prior units. What students are asked to do in the context of simple interest connects to their learning with linear functions. For this lesson, students will be relating previously learned concepts to new concepts and looking for similarities among problems.
 Differentiate Instruction	Simple interest problems can often generate a higher degree of student interest. To capitalize on the student interest, consider extending this lesson by adding in a student choice element. Consider having students be assigned a bank and an account type. Have them research the interest rate information and complete the appropriate calculations to determine what would happen to their money (or debt, depending on the account type) and explain what it means.	
Lesson Notes:		

13.1.1 Warm-Up: Earth's Surface Speed

Component Overview: Students will complete conversions through the context of Earth's surface speed.



13.1.1 Warm-Up: Earth's Surface Speed

The surface of the Earth at the equator moves at a speed of 460 meters per second.

1. Based on your understanding of meters and miles, do you think converting meters per second to miles per second will result in a larger or smaller number?
Smaller
2. Use the conversion 1 mile = 1609.34 meters to change 460 meters per second to miles per second.
0.29 miles per second
3. Use the conversion 1 hour = 3600 seconds to determine the speed of the surface of the Earth in miles per hour.
Answers will vary.
1,044 miles per hour



Pause and Predict

Start with students answering question 1 only. Engage students in a group discussion to see who thinks it will be larger and smaller. Consider counting the votes and charting the count in a centralized location, such as a whiteboard, and then release students to work on questions 2 and 3. Have students come to the board to show their work and answers and reengage students in the conversation for question 1 and see how many predicted correctly.

This protocol encourages both active participation in learning (MTR 1) and analyzing the thinking of others (MTR 4).

13.1.2 Guided Instruction: Simple Interest

Component Overview: Students will learn about the concept of simple interest and connect it to linear functions through two multipart problems.



13.1.2 Guided Instruction: Simple Interest

When money is borrowed or saved, interest is typically accrued. For example, if money is borrowed, the interest accrued is owed in addition to the amount borrowed. If money is saved in an interest-bearing account, the interest accrued is saved in addition to the funds deposited into the account.

Simple interest is calculated by using three quantities — the principal, which could be an amount borrowed or an amount invested; the rate of interest; and the time period it will cover.

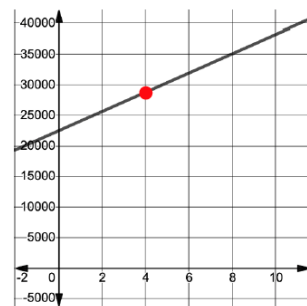
Student loans and mortgages typically use simple interest, whereas savings accounts and credit card debt typically use compound interest.



Simple interest is a method of computing interest. Interest is computed from the (original) principal alone, no matter how much money has accrued so far.

The formula $A = P(1 + rt)$ is used to calculate the final amount owed or saved, where A = final amount, P = principal or original amount, t = number of years, and r = rate of interest per year.

1. Bianca just graduated from college and is ready to buy her first car. She borrows \$22,500 from a bank to make the purchase. The loan she gets has an annual interest rate of 7 percent.



This is not the first time that students have been introduced to the term “simple interest” in the B.E.S.T. standards. In the 7th grade benchmarks, students worked with simple interest. The connection between the concept of simple interest and linear functions, including their graphs and key features, should be the primary focus of instruction to avoid getting caught up in strictly the concept of simple interest in isolation.



For question 1a, students should recognize that this graph is linear.

13.1.2 Guided Instruction: Simple Interest (continued)

- a. What do you notice about the graph?

Answers will vary. I notice that the graph has a positive slope. I notice that the graph is increasing at a constant rate of change, so the graph is linear.

- b. Place a point on the graph that represents the final amount of the loan if Bianca pays the loan off in 4 years.

Answer is shown on graph.

- c. Use the graph to determine how much Bianca will pay for the car if she pays it off in 8 years.

From the graph, the final loan amount is approximately \$35,000.

- d. The equation $A = 22,500(1 + 0.07t)$ models Bianca's loan. Rewrite the equation in slope-intercept form.

$A = 1,575t + 22,500$

- e. Work with your partner to complete the table.

Key Feature	Value	Interpretation in Context
y-intercept	\$22,500	<i>the principal value or the initial amount of Bianca's loan</i>
slope	\$1,575	<i>the amount of interest added to the loan each year</i>

- f. Bianca only wants to pay a total of \$27,500 for the car, including interest on her loan. Use the equation $A = 22,500(1 + 0.07t)$ to determine how long the loan can be to ensure she only pays \$27,500. Round to the nearest hundredth.

3.17 years

A **certificate of deposit** (CD) is a type of savings account offered by banks and credit unions that provides a higher interest rate in exchange for the customer agreeing to leave a lump-sum deposit untouched for a predetermined period of time.



Pay attention to which parts of this question students struggle with to provide additional support where needed. Each piece can determine which area students may be struggling with.

- Question 1a: Students who struggle here may have forgotten the graphs of different function types.
- Question 1b: Students who struggle here may misunderstand ordered pairs and plotting.
- Question 1c: Students who struggle here may have the same struggle as part b or may have reversed the axes in context.
- Question 1d: Students who struggle here may struggle with rearranging formulas (literal equations).



It is highly likely that students are not familiar with a certificate of deposit. Consider spending time showing students what this is and explaining its purpose.

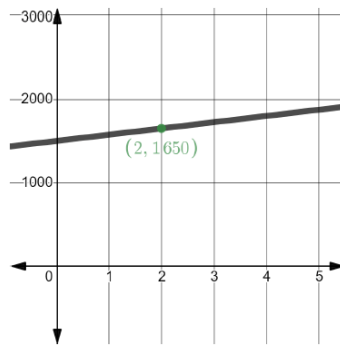
13.1.2 Guided Instruction: Simple Interest (continued)

2. In 2001, Logan's parents had \$1,500 to invest. They chose a certificate of deposit that offered a 5% annual percentage rate (APR).

- a. Work with your partner to write an equation that models his parents' deposit using the formula $A = P(1 + rt)$, where A = final amount, P = principal or original amount, t = number of years, and r = rate of interest per year.

$$A = 1,500(1 + 0.05t)$$

- b. The graph shows the equation modeling Logan's parents' investment and the point (2, 1650).



What does the point (2, 1650) represent?

After 2 years, the total amount with interest added is \$1,650.

The time frames for certificates of deposit are often given in months.

- c. Discuss with your partner if an 18-month CD would have earned less or more money than the point (2, 1650) for Logan's parents. Explain how you determined your answer.

An 18-month CD would earn less money because 18 months is equivalent to 1.5 years, and the graph shows that the output value at 1.5 years is less than the output value at 2 years.



For question 2, students will need a partner. Consider leveraging a Think-Pair-Share protocol when doing this question. Begin by providing students the context and then releasing students to work on part a on their own. Once time is called, have students work with a partner to compare their answers and work together to come to consensus if their answers are not the same. Bring students back together to review the answer to part a before moving to part b.



Consider utilizing inquiry to extend student thinking while reviewing their responses from Think-Pair-Share.

- 18 is greater than 2. Why would Logan have more money when the x -value is 2 rather than 18?
- What would this look like on the graph? Show your reasoning.
- Will the value always be greater as years increase? How do you know? Give an example or counterexample.

13.1.2 Guided Instruction: Simple Interest (continued)

- d. For what length of time, in months, would his parents need a CD if their goal was to have \$1,750?

40 months

- e. Complete the table for simple interest relationships.

Key Feature	Description
End Behavior	As $x \rightarrow \infty$, $y \rightarrow \infty$.
Increasing/Decreasing	As x increases, $y \rightarrow$ <i>increases</i> .

- f. Discuss with your partner how the key features of end behavior and intervals where the function is increasing/decreasing help to explain the relationship between simple interest and linear growth. Summarize your discussion.

Answers will vary. As the amount of interest increases, the total account balance increases. As the amount of the interest decreases, the total account balance decreases.

- g. What other key feature is useful to explain that simple interest is modeled by linear growth?

slope



The goal is to review these concepts in this “bring it together” unit. The questions included here should not take much time to answer individually, as they were covered in prior units. One of the goals is to spiral this content and make connections across units and benchmarks.

This lesson is meant to make connections between simple interest and linear growth so that when students are reviewing the key features of simple interest graphs, they are also making the connection to linear graphs.

13.1.3 Guided Practice: Calculating Simple Interest

Component Overview: Students will solve a problem involving calculating simple interest, including identifying and interpreting two key features in the provided context.



13.1.3 Guided Practice: Calculating Simple Interest

1. Randall is a pro fisherman. He bought a new boat for \$77,500 using a loan with a simple interest rate of 3.75%.

- a. Write an equation that models Randall's loan.

$$A = 77,500(1 + 0.0375t)$$

- b. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept	\$77,500	the initial amount of the boat
slope	\$2,906.25	the amount of interest added each year

- c. What is the total Randall will have to pay if he pays the loan off in 4 years?

\$89,125

- d. How many years did it take Randall to pay off the loan if the total he paid was \$100,750?

8 years



Consider releasing students to work on this task with a partner or leverage a Think-Pair-Share protocol where they can try it on their own and then discuss their answers and work with a partner.

Once time is up, review the answers with the class, including the rationale for each answer, as they will need any clarification necessary prior to moving into the next component, where they will work independently.

13.1.4 Your Turn!

Component Overview: Students will work through a simple interest problem looking for both the amount in the account and time it takes to reach a specified amount.



13.1.4 Your Turn!

1. When Valentina was born, her grandmother put \$2,500 into a savings account for her. The savings account had a 2.25% APR. Each year on her birthday, her grandmother told her how much money was in the account.

a. Complete the table. Show your work.

Valentina's Age	Total Amount
7	$A = 2,500(1 + 0.0225t)$ $A = 2,500(1 + 0.0225(7))$ $A = 2,500(1 + 0.1575)$ $A = 2,500(1.1575)$ $A = 2,893.75$
18	$A = 2,500(1 + 0.0225t)$ $A = 2,500(1 + 0.0225(18))$ $A = 2,500(1 + 0.405)$ $A = 2,500(1.405)$ $A = 3,512.50$

- b. How old will Valentina be when the account has reached \$4,000?
26.6 years old



This component focuses primarily on the simple interest component of the lesson. Allow students to work on this question independently to determine which students may need additional support or what misconceptions may still exist.

Students may make calculation errors when determining the total amount. Consider allowing students the opportunity to use calculators when doing their computations.



How is the process you went through different in question 1a compared to question 1b?

13.1.5 Wrap-Up: Reflection

Component Overview: Students will complete a **SELFIE** reflection to determine which box best represents what they did today based on six statements.



13.1.5 Wrap-Up: Reflection

1. Take a **SELFIE** by checking the box that best matches what you did today.

Answers will vary.

Showed my work	☐	☐
Explained my answers	☐	☐
Lots of math vocabulary used	☐	☐
Found solutions	☐	☐
I persevered through the problem	☐	☐
Eliminated careless errors	☐	☐



Money in the Bank Extension

Consider leveraging real-world bank contexts to extend learning from this lesson while preparing students to engage in Lesson 2. Provide students with the following two prompts. Students must choose which they would rather have and justify their choice mathematically.

Would You Rather...?

Prompt 1: You are given a certificate of deposit worth \$10,000 that has an APR of 20% for 10 years.

Prompt 2: You are given \$10,000 in an account that compounds the interest of 2% every month for 10 years.

Leverage their responses in your questioning as students engage with 13.2.2 Collaboration.